Adversarially Robust Face Presentation Attack Detection: Checkpoint Report

Shivam Shelar
Department of Computer Science, San José State University
Project nature: *Build* an adversarially robust face presentation attack detector.

I. Problem Statement and Motivation

Problem. Face presentation attack detection (PAD) decides, from a single RGB face image x, whether the camera sees a live person or a presentation attack (a printed photo, a poster, or a face replayed on a screen). We learn a classifier $f(x) \in [0, 1]$, the probability that x is live, and report the ISO/IEC 30107-3 error rates [1]: APCER (attacks accepted as live), BPCER (live faces rejected), and their mean ACER. The project goal is a detector that keeps low APCER and BPCER not only on clean images but also when an attacker adds a small, bounded perturbation δ ($\|\delta\|_\infty \leq \epsilon$) to a spoof image to make it pass.

Why it matters. Face verification now gates phone unlock, payments, remote identity checks and self-service kiosks, and print or replay spoofs remain the cheapest way to attack it [2]. Learned PAD models are themselves vulnerable to adversarial examples: AdvGen showed that perturbations which survive printing can push spoofs past state-of-the-art PAD systems [3]. A PAD model that is accurate but trivially fooled therefore gives a false sense of security.

Scope changes since the proposal (in blue). (i) We use the print and replay attack types of CelebA-Spoof (photo, poster, A4 print, PC, pad and phone screens) and exclude masks, which a kiosk camera would face rarely and which add a different problem. (ii) Training uses a balanced, subject-capped subset (Sec. II) rather than all 625K images, to fit the free GPU budget. (iii) The attack remains digital (PGD [4]) as the stand-in for AdvGen's physical attack. (iv) Preliminary experiments showed that robustness at the common $8/255$ budget is not achievable here (Sec. IV-E), so the defense now targets small budgets ($\leq 2/255$) and every model is evaluated with a sweep over $\epsilon \in \{1, 2, 4, 8, 16\}/255$. (v) The trajectory regularizer adapted from [5] is kept only as an ablation. (vi) The webcam demo moves to the final milestone.

II. Data Engineering

Origin. CelebA-Spoof [6]: 625,537 images of 10,177 subjects with 43 attributes per image (40 face attributes plus spoof type, illumination and environment) and a face bounding box per image. It is distributed for non-commercial research only; we use the Kaggle copy (`attentionlayer241/celeba-spoof-for-face-antispoofing`, about 753K files, 58.8 GB) and its official intra-test protocol, whose train and test subjects are disjoint. No data is committed to the repository; the README gives the download and preparation commands.

Characteristics. The copy has 8,192 training and 1,004 test subject folders. Labels: live or spoof, spoof type (1–10), illumination (normal, strong, back, dark), environment (indoor, outdoor). Source images have varied resolutions (median [W×H] px from `eda.py`).

Cleaning and preprocessing. The script `data/prepare_celeba_spoof.py` applies five steps: (1) keep only labelled images present on disk and only the six print/replay types; (2) crop a square of 1.5× the annotated face box, because the background (screen bezel, paper edge) carries attack evidence but the full frame is dominated by scene content; (3) *reject* crops that run outside the photo: on our first load, zero-padded borders appeared in 33% of live but 60% of spoof crops, a shortcut a CNN would exploit; after the fix, dark-edge rates fell to 2.3% and 5.3%; (4) take at most 8 images per subject per class, which spreads the same budget over about four times as many people (2,000+ training subjects instead of about 500); (5) resize to 224×224 and normalize with ImageNet statistics; training adds horizontal flips and colour jitter. Sampling is deterministic (sorted listings, seed 42). No hand-made features or PCA are used: the CNN learns directly from pixels, and Sec. III shows simple image statistics do not separate the classes.

Schema sanity and quality. No sample lacked a face box. Besides the padding artifact above, we checked subject leakage ([0] subjects shared between splits; disjoint by construction) and near-duplicates via 16×16 image hashes ([N] groups, [N] across splits, [N] with conflicting labels). Shortcomings for our problem: one dataset only; spoofs are re-captures of celebrity photos; and the test split's spoof-type mix differs from training (Fig. 1), which we return to in Sec. IV.

Split. Table I. Validation subjects are a random 15% of training subjects; test comes from the official test split. Classes are balanced by construction.

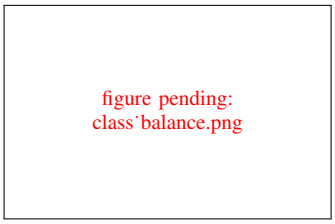
III. Exploratory Data Analysis

All figures are produced by `evaluation/eda.py`.

Class balance (Fig. 1). Equal class counts mean accuracy is not misleading, but because the two error types have different costs we always report APCER and BPCER separately, with

TABLE I
PREPARED FACE CROPS (LIVE / SPOOF).

Split	Crops per class	Subjects	Source
Train	2,550 / 2,550	[n]	CelebA-Spoof train
Val	450 / 450	[n]	held-out train subjects
Test	600 / 600	[n]	CelebA-Spoof test



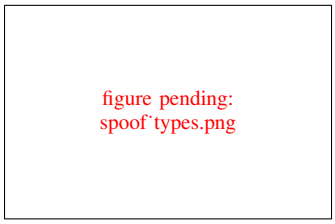


Fig. 1. Class balance and spoof-type mix per split. Classes are balanced, but PC-screen replays are 40.5% of test spoofs against 17.6% of training spoofs, a shift that later explains most baseline errors.

counts. The spoof-type imbalance between train and test motivates per-type error analysis rather than oversampling, since the test mix is fixed by the protocol. **Univariate** (Fig. 2) and **bivariate** (Fig. 3) analyses show [summary from the figures and eda_summary.json correlations].

IV. Baseline and Preliminary Results

A. Setup

ResNet-18 [7] with ImageNet weights and a single output logit, trained with binary cross-entropy, AdamW (lr 10^{-4}), batch 32, 5 epochs; the epoch with the best validation accuracy is kept. Three seeds (0, 1, 2) fix all randomness (`utils/seed.py`); the commit hash is saved with the results.

B. Metrics with counts

Table II. For reference, the pilot run (seed 0, same data) gave APCER 22.5% (135/600), BPCER 2.0% (12/600) and ACER 12.2%, with validation accuracy 99.1% against test accuracy 87.8%; two further pilot runs gave APCER 20.2% and 19.8%.

C. Threshold analysis

We choose the operating point on *validation* (its EER threshold) and apply it unchanged to test, so the test set never tunes the model (Fig. 4). [state the threshold and the resulting APCER/BPCER, and why this point is preferred, e.g. for a kiosk where a false accept is costlier than a retry].

D. Error analysis

Errors concentrate in one attack type (Table III): PC-screen replays account for 128 of the 135 false accepts, while every training fake, PC screens included, is rejected. Failures also cluster in normal lighting (29.5% accepted) and indoor scenes (25.7%), against 0–4% under strong, back or dark light and 3.5% outdoors. Inspecting the accepted PC-screen crops shows sharp, well-lit captures with no visible pixel grid or moiré, while correctly rejected ones show that texture. **Hypothesis:**

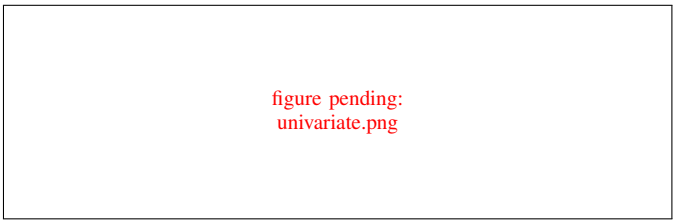
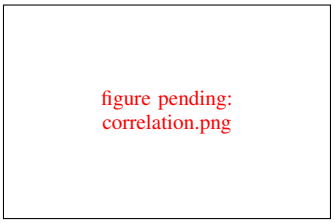


Fig. 2. Univariate image statistics (brightness, contrast, saturation, sharpness, high-frequency energy) as histograms (top) and box plots (bottom), live vs spoof. [one sentence on the overlap seen]



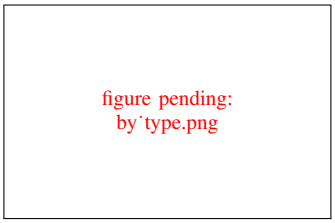


Fig. 3. Left: correlation of the statistics with each other and with the label. Right: sharpness and high-frequency energy by presentation type. [one sentence: which types sit close to live]

the baseline relies on fine screen texture, which disappears in high-quality captures; the test split contains many more of these than training, which also explains the gap between validation (99%) and test (88%) accuracy. [add the false-accept grid figure if space allows]

E. Preliminary robustness results

A 10-step PGD attack flips 100% of the baseline's test fakes at every budget from $1/255$ to $16/255$. Adversarial training at $2/255$ lowers this: among fakes the hardened model rejects without attack, a $2/255$ attack flips 29% (11% at $1/255$), reproduced across two training schemes, at the cost of a higher clean APCER (about 42% vs 20%), concentrated on screen replays. Training at $8/255$ with up-weighted fakes appeared to give 2% attack success, but the budget sweep exposed it: smaller attacks still succeed 98% of the time, which is impossible for a truly robust model and indicates gradient masking [8].

Reproducibility. All numbers in this section come from `notebooks/kaggle_checkpoint.ipynb` at commit [hash]; the repository README maps each script to its section.

References

[1] ISO/IEC, "ISO/IEC 30107-3: Information technology – biometric presentation attack detection – part 3: Testing and reporting," 2017.

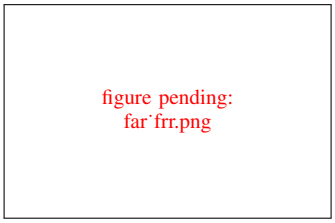
[2] Z. Yu, Y. Qin, X. Li, C. Zhao, Z. Lei, and G. Zhao, "Deep learning for face anti-spoofing: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 5, 2023.

[3] S. A. Patnaik, S. Chansoriya, A. K. Jain, and A. M. Namboodiri, "AdvGen: Physical adversarial attack on face presentation attack detection systems," in *IEEE International Joint Conference on Biometrics (IJCB)*, 2023.

[4] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *International Conference on Learning Representations (ICLR)*, 2018.

TABLE II
BASELINE ON THE TEST SET (600 LIVE, 600 SPOOF). MEAN ± STD OVER 3 SEEDS; COUNTS FOR SEED 0.

	Default (p=0.5)	Operating point
Threshold	0.500	[val EER thr.]
APCER	[% (n/600)]	[]
BPCER	[% (n/600)]	[]
ACER	[]	[]
EER (val / test)	[]	
AUC (val / test)	[]	



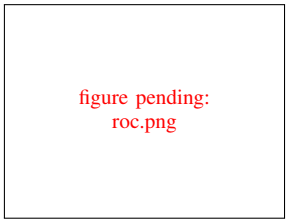


Fig. 4. Left: APCER (FAR) and BPCER (FRR) against the threshold on validation and test, with the chosen operating point. Right: ROC curves. [one sentence on the val–test gap visible here]

[5] Y. Sun, Y. Liu, X. Liu, Y. Li, and W.-S. Chu, "Rethinking domain generalization for face anti-spoofing: Separability and alignment," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[6] Y. Zhang, Z. Yin, Y. Li, G. Yin, J. Yan, J. Shao, and Z. Liu, "CelebA-Spoof: Large-scale face anti-spoofing dataset with rich annotations," in *European Conference on Computer Vision (ECCV)*, 2020.

[7] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[8] A. Athalye, N. Carlini, and D. Wagner, "Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples," in *International Conference on Machine Learning (ICML)*, 2018.

TABLE III
TEST SPOOFS ACCEPTED AS LIVE BY SPOOF TYPE (PILOT, SEED 0, p=0.5).

Spoof type	Accepted / total	Rate
PC screen	128 / 243	52.7%
Poster	3 / 16	18.8%
Photo	3 / 76	3.9%
Phone screen	1 / 202	0.5%
A4 print	0 / 39	0.0%
Pad screen	0 / 24	0.0%
All	135 / 600	22.5%